The Renormalization Group idea (RG)

At the critical point $G^c(\vec{x}) = \langle m(\vec{x})\, m(0)\rangle - \langle m^2\rangle \sim \dfrac{1}{x^{d-2+\eta}}$

(η = anomalous exponent, $\eta = 0$ in simple LG analysis)

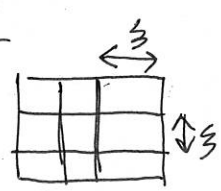
<u>Scale invariance</u>: $G^c(\lambda \vec{x}) = \dfrac{1}{\lambda^{d-2+\eta}\, x^{d-2+\eta}} = \lambda^{2-d-\eta}\, G^c(\vec{x})$

Thus, critical system at scale λL looks the same as that at scale L, upto "intensity" rescaling factor $\lambda^{2-d-\eta}$.
That is, the system is self-similar — this is property of fractal geometry.

The RG idea is to investigate behavior of the system under rescaling (dilation) transformations.

<u>Widom</u> suggested that free energy should be homogeneous function of t $\left(t = \frac{T-T_c}{T_c}\right)$ and h (magnetic field)

$$F(t,h) \xrightarrow[L \to \lambda L]{} \lambda^{-d} F(t', h') \quad (d = \text{dimensionality})$$



being extensive ($\sim$ volume), free energy F must scale as volume L^d.

$\Rightarrow F \sim \left(\dfrac{L}{\xi}\right)^d$ where ξ = correlation length

free energy density $f = \dfrac{F}{L^d} \sim \xi^{-d}$

hence, under scaling $L \to \lambda L$, $\xi \to \lambda \xi$,

$\Rightarrow \underline{f \to \lambda^{-d} f}$

Under $L \to \lambda L$, $t' = \lambda^{y_t} t$, $h' = \lambda^{y_h} h$

where y_t, y_h are exponents (need to be determined)

$$\Rightarrow f(t,h) = \lambda^{-d} f(\lambda^{y_t} t, \lambda^{y_h} h)$$

Now, let us <u>choose</u> λ such that $\lambda^{y_t} t = 1$

$$\Rightarrow \lambda = t^{-\frac{1}{y_t}}$$

$$\Rightarrow f(t,h) = t^{\frac{d}{y_t}} f(1, t^{-\frac{y_h}{y_t}} h) = t^{\frac{d}{y_t}} g\left(\frac{h}{t^{y_h/y_t}}\right)$$

From this scale-invariance, a number of relations between the critical exponents follow.

* Heat capacity

$$C_{sing} \sim -\frac{\partial^2 f}{\partial t^2}$$

Say, $t > 0$:

$$\frac{\partial f}{\partial t} = \frac{d}{y_t} t^{\frac{d}{y_t}-1} g\left(\frac{h}{t^{y_h/y_t}}\right) + t^{\frac{d}{y_t}} g'\left(\frac{h}{t^{y_h/y_t}}\right) \frac{\partial}{\partial t}\left(\frac{h}{t^{y_h/y_t}}\right)$$

$$= \frac{d}{y_t} t^{\frac{d}{y_t}-1} g + t^{\frac{d}{y_t}} g' h \left(-\frac{y_h}{y_t}\right) t^{-\frac{y_h}{y_t}-1} =$$

$$= t^{\frac{d}{y_t}-1}\left(\frac{d}{y_t} g - \frac{y_h}{y_t} \frac{h}{t^{y_h/y_t}} g'\right) = t^{\frac{d}{y_t}-1} \tilde{g}\left(\frac{h}{t^{y_h/y_t}}\right)$$

another homogeneous function

$$\Rightarrow C_{sing} \sim -\frac{\partial^2 f}{\partial t^2} \sim t^{\frac{d}{y_t}-2} \tilde{\tilde{g}}\left(\frac{h}{t^{y_h/y_t}}\right)$$

Experimentally, $C_{sing} \sim |t|^{-\alpha} \Rightarrow \boxed{\alpha = 2 - \frac{d}{y_t}}$

Key idea: correlation length $\xi = \xi(t, h)$ is the <u>only</u> spatial scale in the system.

Behavior at distances much smaller ξ is not important $\Rightarrow$ <u>coarse-grain</u> system to "blocks" of size ξ.

(Kadanoff, Migdal, Pokrovsky, Popov, Wilson, Fisher...)

(1) The microscopic scale a (lattice spacing between spins, for example)

Decrease the resolution by <u>coarse-graining</u> the field

$$\boxed{a \to ba}$$

$$\varphi(\vec{x}) = \frac{1}{b^d} \int_{(\text{cell at } \vec{x})} d\vec{x}' \, \varphi(\vec{x}')$$

(2) To restore the resolution a, <u>rescale</u> the length scale

$$\vec{x}_{new} = \frac{\vec{x}_{old}}{b}$$

(3) <u>Renormalize</u>: rescaled field $\varphi(\vec{x})$ is characterized, in general, by a different "contrast" (magnitude of variations) from the original one.

This can be remedied by a change of contrast by a factor ζ, by defining a <u>renormalized</u> field

$$\varphi_{new}(\vec{x}_{new}) = \frac{1}{\zeta b^d} \int_{(\text{cell at } b\vec{x}_{new})} d\vec{x}' \, \varphi(\vec{x}').$$

In this way we generate new "hamiltonian" (action) which is expressed in terms of φ_{new} and <u>new</u> coupling <u>constants</u>

$$t_{new} = t_b(t_{old}, h_{old})$$

function

$$h_{new} = h_b(t_{old}, h_{old})$$

$$\begin{cases} t_b(t,h) = A(b)\,t + \cancel{B(b)}\,h + \dots \\ h_b(t,h) = \cancel{C(b)}\,t + D(b)\,h + \dots \end{cases}$$

regular expansion because transformations on short scales cannot cause singularities!

Observe: (1) no constant terms in the expansion because at the critical point $t=0$, $h=0$

$\xi = \infty \Rightarrow$ renormalization must result in the same Hamiltonian (fixed point!)

(2) Symmetry $\varphi \to -\varphi$, $h \to -h$, $t \to t$ leaves Ham. the same

$\Rightarrow B = C = 0$

$$\begin{cases} t_b(t,h) = A(b)\,t + \dots \\ h_b(t,h) = D(b)\,h + \dots \end{cases}$$

$$A(b=1) = D(b=1) = 1.$$

Semi-group: rescaling by b_1, and then by b_2, gives rescaling by $b_1 b_2$.

$$\Rightarrow t_b(b_1 b_2) = A(b_1)\,A(b_2)\,t = A(b_1 b_2)\,t$$

$$\Rightarrow A(b) = b^{y_t} \Rightarrow \begin{cases} t' = t_b = b^{y_t}\,t + \dots \\ h' = h_b = b^{y_h}\,h + \dots \end{cases}$$

must be!

Formal RG

(1) $$S = \int d^d\vec{x} \left\{ \frac{1}{2} t \varphi^2 + \frac{1}{4} u \varphi^4 + \frac{1}{6} v \varphi^6 + \ldots + \right.$$
$$\left. + \frac{1}{2} \rho (\nabla\varphi)^2 + \frac{1}{2} \omega (\nabla^2\varphi)^2 + \ldots - h\varphi \right\}$$

Thus, a particular physical system is characterized by a point in the parameter space $(t, u, v, \ldots, \rho, \omega, \ldots h) = P$

(2) $$\varphi'(\vec{x}') = \frac{1}{\xi b^d} \int_{\substack{\text{cell of size } ba \\ \text{centered at } bx'}} d^d\vec{x}\ \varphi(\vec{x}).$$

(3) The rescaled action must be

$$S'(\varphi'(x')) = f_b + \int d^d\vec{x}' \left\{ \frac{1}{2} t' (\varphi')^2 + \frac{1}{4} u' (\varphi')^4 + \right.$$
$$\left. + \frac{1}{6} v' (\varphi')^6 + \ldots + \frac{1}{2} \rho' (\nabla'\varphi')^2 \ldots - h'\varphi' \right\}$$

$$\begin{cases} t' = t_b(t, u \ldots) \\ h' = h_b(t, u \ldots) \\ u' = u_b(t, u \ldots) \text{ etc.} \end{cases}$$

In parameter space, we have mapping
$$P' = R_b P$$

(4) Statistically self-similar configuration must correspond to a <u>fixed point</u> P^* of the transformation such that $R_b P^* = P^*$.

Since $\xi(P) = b\, \xi(R_b P)$, at the fixed point
$\xi^* = \infty$ (the other possibility, $\xi^* = 0$ describes completely uncorrelated system)
$\Rightarrow t^* = 0$

(5) Linearize recursion relations: under RG point $P^* + \delta P$ transforms as

$$P^*_\alpha + \delta P_\alpha = P^*_\alpha + (R_b)_{\alpha\beta}\, \delta P_\beta + \dots$$

Matrix $(R_b)_{\alpha\beta}$ is characterized by eigenvectors O_i and eigenvalues λ_i

$$\hat{R}_b\, O_i = \lambda_i(b)\, O_i$$

Since $R_{b_1} R_{b_2} O_i = \lambda_i(b_1)\, \lambda_i(b_2)\, O_i = R_{b_1 b_2} O_i =$

$$= \lambda_i(b_1 b_2)\, O_i$$

$$\Rightarrow \lambda_i(b) = b^{y_i}$$

y_i = anomalous dimensions (exponents)

Near the fixed point

$$S = S^* + \sum_i g_i O_i$$

$$S' = S^* + \sum_i g_i\, b^{y_i} O_i$$

- $y_i > 0$, g_i increases under scaling, O_i = relevant operator
- $y_i < 0$, g_i decreases

Perturbative RG (Wilson)

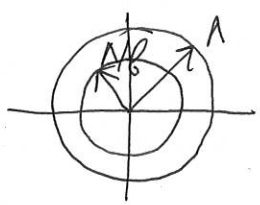
$$Z = \int D\varphi \, \exp - S_{LG}$$

$$S_{LG} = \int d\vec{x} \; \frac{1}{2}\rho (\nabla\varphi)^2 + \frac{1}{2} t \varphi^2 + \frac{1}{4} u \varphi^4 + \ldots = S_0 + S_u =$$

$$= \int \frac{d\vec{q}}{(2\pi)^d} \left(\frac{1}{2} t + \frac{1}{2}\rho q^2 + \ldots \right) |\varphi(\vec{q})|^2 \qquad (S_0)$$

$$+ u \int \frac{d\vec{q}_1 d\vec{q}_2 d\vec{q}_3}{(2\pi)^{3d}} \; \varphi_\alpha(\vec{q}_1)\, \varphi_\alpha(\vec{q}_2)\, \varphi_\beta(\vec{q}_3)\, \varphi_\beta(-\vec{q}_1 - \vec{q}_2 - \vec{q}_3) \qquad (S_u)$$

(1) <u>Coarse grain</u>: eliminating fields at scales $a < x < ba$ is similar to removing Fourier modes with wavenumbers $\frac{\Lambda}{b} < q < \Lambda$



$$\Rightarrow \{\vec{\varphi}(\vec{q})\} = \underset{0 \le q \le \frac{\Lambda}{b}}{\{\vec{\varphi}_<\}} \otimes \underset{\frac{\Lambda}{b} < q < \Lambda}{\{\vec{\varphi}_>\}}$$

$$\vec{\varphi}(\vec{x}) = \frac{1}{\sqrt{V}} \sum_{\vec{q}} e^{i\vec{q}\cdot\vec{x}} \vec{\varphi}(\vec{q}) =$$

$$= \underbrace{\frac{1}{\sqrt{V}} \sum_{\{0<q<\frac{\Lambda}{b}\}}{}' e^{i\vec{q}\cdot\vec{x}} \vec{\varphi}(\vec{q})}_{\vec{\varphi}_<(\vec{x}) \text{ (slow modes)}} + \underbrace{\frac{1}{\sqrt{V}} \sum_{\{\frac{\Lambda}{b}<q<\Lambda\}}{}'' e^{i\vec{q}\cdot\vec{x}} \vec{\varphi}(\vec{q})}_{\vec{\varphi}_>(\vec{x}) \text{ (fast modes)}}$$

$$Z = \int D\varphi(\vec{q})\, e^{-S[\varphi(\vec{q})]} = \int D\varphi_<(\vec{q}) \int D\varphi_>(q)\, e^{-S[\varphi_< + \varphi_>]}$$

Gaussian model S_0 $(u=0)$

$$S_0 = \sum_q \left(\frac{1}{2} t + \frac{1}{2} p \vec{q}^{\,2}\right) \vec{\varphi}(\vec{q}) \vec{\varphi}(-\vec{q}) =$$

$$= \sum_{\underline{0 \le q \le \frac{\Lambda}{b}}}{}' \left(\frac{1}{2} t + \frac{1}{2} p \vec{q}^{\,2}\right) \vec{\varphi}_<(\vec{q}) \vec{\varphi}_<(-\vec{q})$$

$$+ \sum_{\underline{\frac{\Lambda}{b} \le q \le \Lambda}}{}'' \left(\frac{1}{2} t + \frac{1}{2} p \vec{q}^{\,2}\right) \vec{\varphi}_>(\vec{q}) \vec{\varphi}_>(-\vec{q})$$

On the gaussian level, no mixing between slow and fast components $\varphi_<$ and $\varphi_>$.

(2) <u>Rescale</u> $\vec{\varphi}_<(\vec{q})$ ~~has~~ ~~the~~ q restricted by $\frac{\Lambda}{b}$.

Rescaling $\boxed{\vec{x}' = \frac{\vec{x}}{b}} \Leftrightarrow \boxed{\vec{q}' = b\,\vec{q}}$

$\Rightarrow$ cut-off changes $\frac{\Lambda}{b} \to b \frac{\Lambda}{b} = \Lambda$

$$Z = \int D\varphi_>(\vec{q})\, e^{-\sum_q{}'' \frac{1}{2}(t + p\vec{q}^{\,2}) \vec{\varphi}_>(\vec{q}) \vec{\varphi}_>(-\vec{q})} \cdot$$

$$\cdot \int D\varphi_<(\vec{q}) \exp - \sum_q{}' \frac{1}{2}(t + p\vec{q}^{\,2}) \vec{\varphi}_<(q) \vec{\varphi}_<(-q)$$

$$= \exp\left\{-\frac{1}{2} \sum_q{}'' \ln(pq^2 + t)\right\} \cdot$$ from integrating out $\varphi_>$

$$\cdot \int D\varphi_<(\vec{q}') \exp - \frac{1}{2} \underbrace{\int_0^\Lambda \frac{d^d \vec{q}'}{(2\pi)^d}}_{\int_0^{\Lambda/b} \frac{d\vec{q}}{(2\pi)^d}} b^{-d} \frac{1}{2}\left(t + p\, \underbrace{b^{-2} (\vec{q}')^2}_{(\vec{q})^2}\right) \left(\vec{\varphi}_<(\vec{q}')\right)^2$$

(3) <u>Renormalize</u> : $\vec{\varphi}'(\vec{x}') = \dfrac{\vec{\varphi}_<(\vec{x}')}{\zeta}$

Here $\vec{\varphi}_<(\vec{x}) = \dfrac{1}{\sqrt{V}} \sum_{q \le \frac{\Lambda}{b}}' e^{i\vec{q}\cdot\vec{x}}\, \vec{\varphi}_<(\vec{q}) = \begin{pmatrix} \vec{q} = \frac{\vec{q}'}{b} \\ \vec{x} = b\vec{x}' \end{pmatrix}$

$$= \frac{1}{\sqrt{V}} \sum_{|\vec{q}'| \le \Lambda} e^{i\frac{\vec{q}'}{b}\cdot b\vec{x}'}\, \vec{\varphi}_<\left(\frac{\vec{q}'}{b}\right) = \vec{\varphi}_<(\vec{x}')$$

Thus, renormalization $\vec{\varphi}'(\vec{x}') = \vec{\varphi}_<(\vec{x}')/\zeta$
is equivalent to
$$\vec{\varphi}'(\vec{q}') = \vec{\varphi}_<\left(\frac{\vec{q}'}{b}\right)/Z$$

Or, $\vec{\varphi}_<\left(\frac{\vec{q}'}{b}\right) = Z\,\vec{\varphi}'(\vec{q}')$

$$\Rightarrow Z = \exp\left[-\frac{1}{2}\sum_{\frac{\Lambda}{b} \le q \le \Lambda}'' \ln(\rho q^2 + t)\right] \cdot$$

$$\cdot \int D\vec{\varphi}'(\vec{q}')\, \exp -\frac{1}{2} \int_0^\Lambda \frac{d\vec{q}'}{(2\pi)^d}\, b^{-d}\, Z^2 \frac{1}{2}\left(t + \rho b^2 (\vec{q}')^2\right) \cdot$$
$$\cdot\, \vec{\varphi}'(\vec{q}')\, \vec{\varphi}'(-\vec{q}')$$

Thus $\begin{cases} t' = Z^2 b^{-d}\, t \\ \rho' = Z^2 b^{-d-2}\, \rho \end{cases}$

To make S_0^* scale invariant at the critical point $t=0$ we require $\rho' = \rho \Rightarrow Z = b^{1+d/2}$

$$\Rightarrow \begin{cases} t' = \left(b^{1+\frac{d}{2}}\right)^2 b^{-d}\, t = b^2 t \\ \rho' = \rho \end{cases} \Rightarrow \underline{y_t = 2}$$

$\boxed{t=0}$

$$S_0^* = \int \frac{d^d\vec{q}}{(2\pi)^d}\, \frac{1}{2}\rho\, \vec{q}^{\,2}\, \vec{\varphi}(\vec{q})\, \vec{\varphi}(-\vec{q}) =$$

$$= \int d^d\vec{x}\, \frac{1}{2}\rho \left(\vec{\nabla}\varphi(\vec{x})\right)^2$$

<u>Observe:</u> $\vec{x} \to \vec{x}' = \frac{\vec{x}}{b}$

$$\vec{\varphi}(x) \to \xi\, \vec{\varphi}'(x')$$

$$\int d^d\vec{x}\, \frac{1}{2}\rho\left(\vec{\nabla}\vec{\varphi}\right)^2 = \int d^d\vec{x}'\, b^d\, \frac{1}{2}\rho\left(\frac{1}{b}\vec{\nabla}'\xi\,\vec{\varphi}'(\vec{x}')\right)^2$$

$$\left[\frac{\partial}{\partial x} = \frac{\partial}{\partial x'}\frac{\partial x'}{\partial x} = \frac{1}{b}\frac{\partial}{\partial x'}\right]$$

$$\Rightarrow \rho \to \underbrace{\rho' = b^{d-2}\xi^2\rho}_{\text{invariant if } \xi = b^{1-\frac{d}{2}}}$$

This means $u_n \int d\vec{x}\, (\vec{\varphi}(x))^n \to u_n\, b^d \xi^n \int d\vec{x}'\left(\vec{\varphi}'(\vec{x}')\right)^n$

$$u_n' = b^d\left(b^{1-\frac{d}{2}}\right)^n u_n = b^{n-d\left(\frac{n}{2}-1\right)} u_n$$

Thus $y_n = n - d\left(\frac{n}{2}-1\right)$

$n=1$: $y_1 = 1 - d\left(\frac{1}{2}-1\right) = 1 + \frac{1}{2}d$

$n=2$: $y_2 = 2 - d(1-1) = 2$

$n=4$: $y_4 = 4 - d\left(\frac{4}{2}-1\right) = 4-d$ $\left[\begin{array}{l} y_4 < 0 \text{ for } d>4 \\ y_4 > 0 \text{ for } d<4 \end{array}\right.$

Differential formulation: let $b = 1 + \delta l$

$$t' = b^2 t$$

RHS: $t' = t(b) = t(1+\delta l) = t + \delta l \frac{dt}{dl} + O(\delta l^2)$

LHS: $b^2 t = (1 + 2\delta l) t$

$$\Rightarrow \quad t + \delta l \frac{dt}{dl} = (1 + 2\delta l) t = t + 2\delta l \, t$$

$$\Rightarrow \frac{dt}{dl} = 2t \quad \Rightarrow \quad t(l) = t(l_0) e^{2(l - l_0)}$$

Then $u_n' = b^{y_n} u_n$

$$u_n + \delta l \frac{du_n}{dl} = (1 + y_n \delta l) u_n = u_n + y_n u_n \delta l$$

$$\Rightarrow \frac{du_n}{dl} = y_n u_n \Rightarrow u_n(l) = u_n(l_0) e^{y_n (l - l_0)}$$

At the gaussian level $y_n = n - d\left(\frac{n}{2} - 1\right)$.

<u>Next</u> add $u \int dx (\varphi)^4$ term and treat it <u>perturbatively</u>